

5(a)	any two from: 1 bacteria, have <u>70S</u> ribosomes and humans have <u>80S</u> ribosomes / <u>only</u> have 70S ribosomes ; 2 antibiotic, cannot bind to 80S / human ribosomes or can only bind to 70S / bacterial ribosomes ; 3 antibiotic, binding site / target site, not present on 80S ribosomes / only present on 70S ribosomes ; A ref. to complementary / specific – alternative to second part of the MP 4 antibiotic cannot enter mitochondria to reach 70S ribosomes (in humans) ; 5 AVP ; different combinations of proteins in ribosomes / different rRNAs in ribosomes	2
5(b)	any one from: 1 penicillin acts on cell walls <u>and</u> human cells have no, peptidoglycans / cell walls / enzyme(s) that make cell walls; A murein <i>penicillin</i> 2 prevents formation of peptide / cross, bridges / linkages, (between peptidoglycans) in (bacterial) cell wall ; 3 prevents repair of, gaps / holes / AW, in (bacterial) cell walls caused by autolysins ; I ref. to penicillinase	1
5(c)	any one from: <i>rifampicin</i> 1 blocks / fits into / binds to / competes for, the active site of, RNA polymerase / enzyme ; 2 binds, to allosteric site / site away from active site, to, change shape of active site / prevent it forming phosphodiester bonds ; 3 attaches to DNA template blocking RNA polymerase moving along it ; 4 can be incorporated into RNA but no nucleotide can be added to it ; 5 AVP ; e.g. attaches to the promoter to stop RNA polymerase, binding to DNA / moving along DNA / AW	1